Contents

RFID Attendance System Documentation

1. Introduction

1.1 Background

The RFID Attendance System is designed to replace traditional attendance methods in educational institutions which usually involves taking attendance manually on paper. The traditional method is prone to errors, is time consuming and allows proxy. Hence the RFID attendance system replaces this method to make taking attendance faster, easier, accurate and more secure.

1.2 Purpose of the Project

The main purpose is to automate taking attendance with the help of RFID Cards and an ESP32 microcontroller. The attendance should be recorded in real-time and uploaded to the cloud (Google Sheets) for easy access.

1.3 Scope and Objectives

The scope of this system includes:

- Scanning RFID Cards
- Verifying Student Identity
- Detecting and preventing duplicate attendance
- Storing attendance online

The objectives of this system are:

- To remove manual work
- Ensure accuracy in records
- Make attendance available on the cloud

1.4 Target Audience

The target audience includes students as their attendance is recorded automatically, teachers who can monitor attendance and no longer must do manual work and educational institutions that can maintain records on the cloud with ease of access.

2. Features and Functionality

2.1 RFID-Based Identification

Each student is assigned an RFID card which contains a unique identifier called the UID. When the card is scanned, the UID is extracted and used to identify the student.

2.2 Student Data Mapping

Each student's details like roll number, name and class/section are mapped to their corresponding card's UID and this ensures correct identification.

2.3 Attendance Verification

When a valid RFID card is scanned, the system verifies the student and marks their attendance at the exact date and time on Google Sheets. For confirmation, the LCD displays the student's name and verification status.

2.4 Duplicate Scan Detection

The system prevents multiple attempts/duplicate entries by keeping track of the UIDs scanned. If any of the same cards are scanned again, they are detected as duplicate and ignored.

2.5 Audio and Visual Feedback

A buzzer and 2 LEDs are used to provide audio and visual feedback to users. A green LED and short beep from the buzzer indicate a valid scan while a red LED and double beep indicate a duplicate scan.

2.6 Cloud Integration

The system uploads attendance to Google Sheets allowing real time monitoring and record keeping.

3. Implementation Details

3.1 Hardware Components

- ESP32 Microcontroller + Power Supply (USB Cable)
- RFID Reader (RC522)
- RFID Cards
- Jumper Wires
- Breadboard
- OLED Display
- Buzzer
- LEDs

3.2 Software Tools

- VS Code
- PlatformIO Extension
- Google Apps Script
- Google Sheets

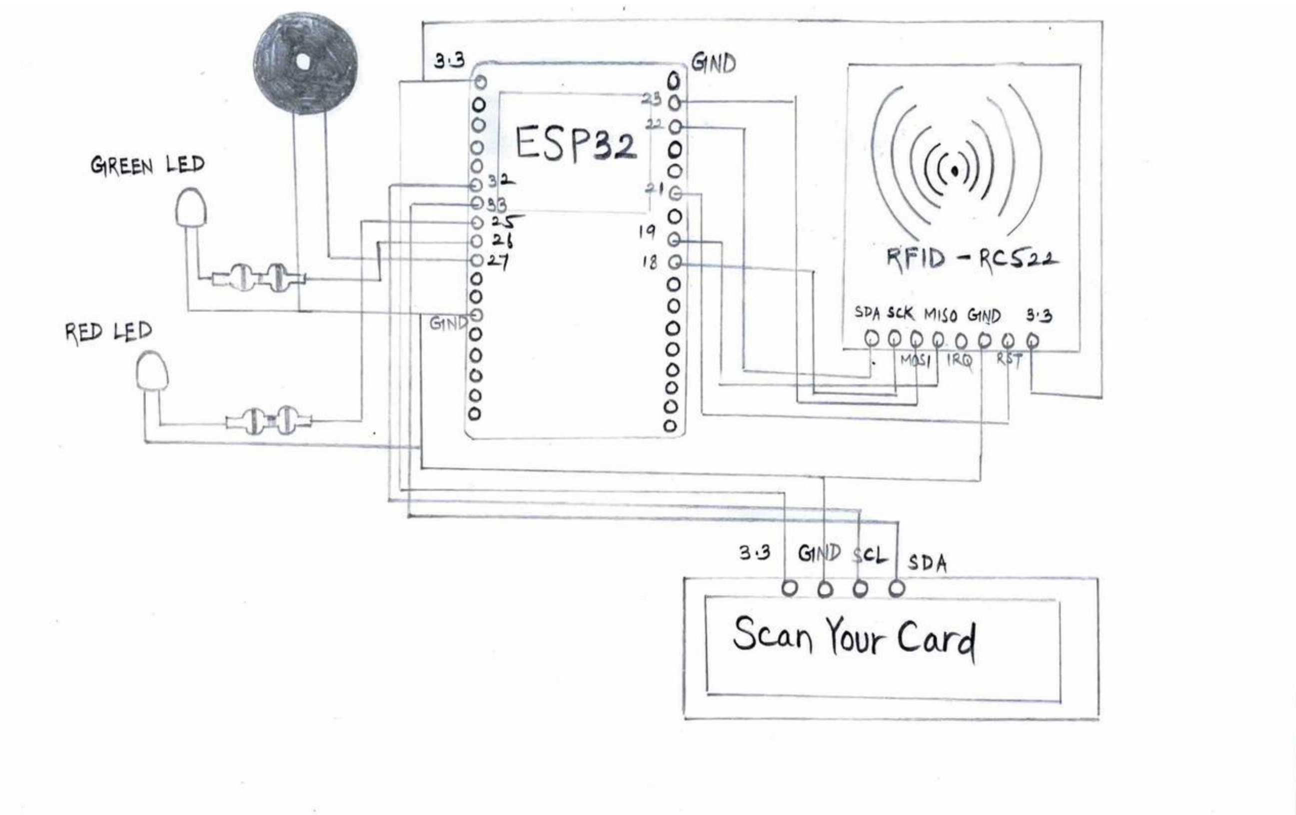
3.3 Libraries Used

- MFRC522
- WiFi
- HTTPClient
- LiquidCrystal_I2C
- time.h

4. System Workflow

RFID Card ----scanned----> RFID Reader ----reads UID----> ESP32 Microcontroller ----(UID Verification, Duplicate Check, Attendance Record Prepared, Feedback Displayed) Attendance Record via Wi-Fi---> Google Apps Script Web App ----> Google Sheets (Cloud Dashboard)

5. Assembly & Circuit Diagram



RFID RC522	Pin / Connection
SDA	22
SCK	18
MOSI	23
MISO	19
IRQ	Not Connected
GND	GND
RST	21
3.3	3.3

LEDs	Pin / Connection
Green Led	26
Red Led	25

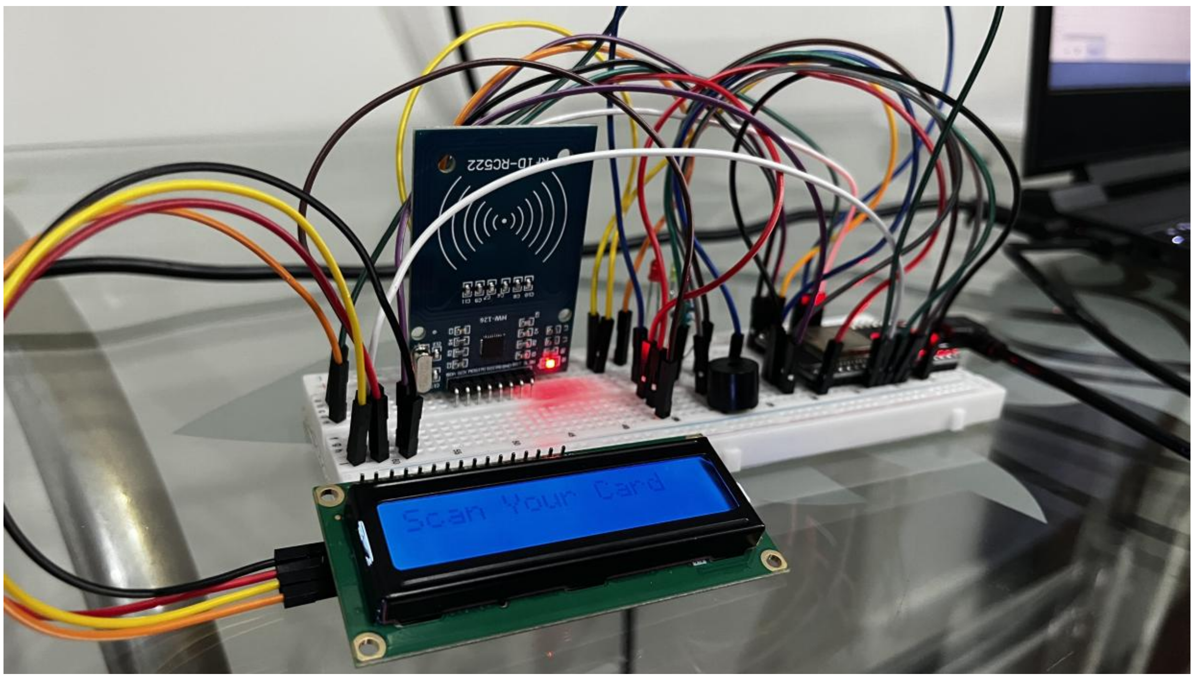
Buzzer	Pin / Connection
Buzzer	27

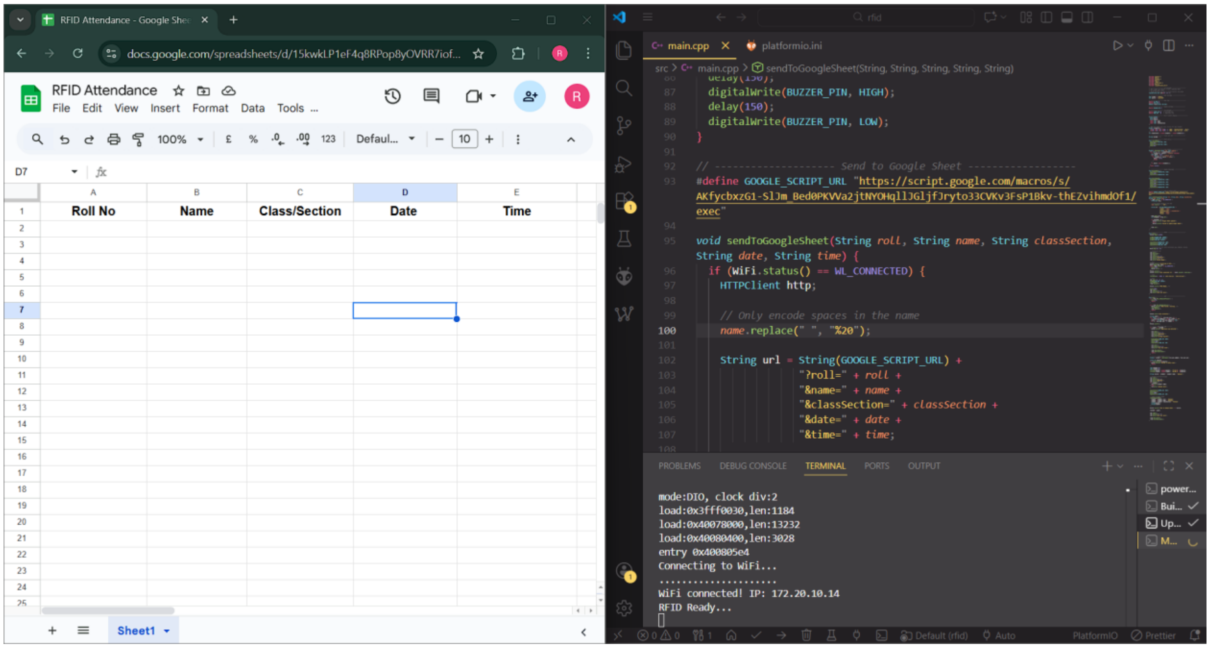
LCD (I2C)	Pin / Connection
SDA	33
SCL	32
VCC	3.3V
GND	GND

6. Testing and Results

6.1 Initial Setup

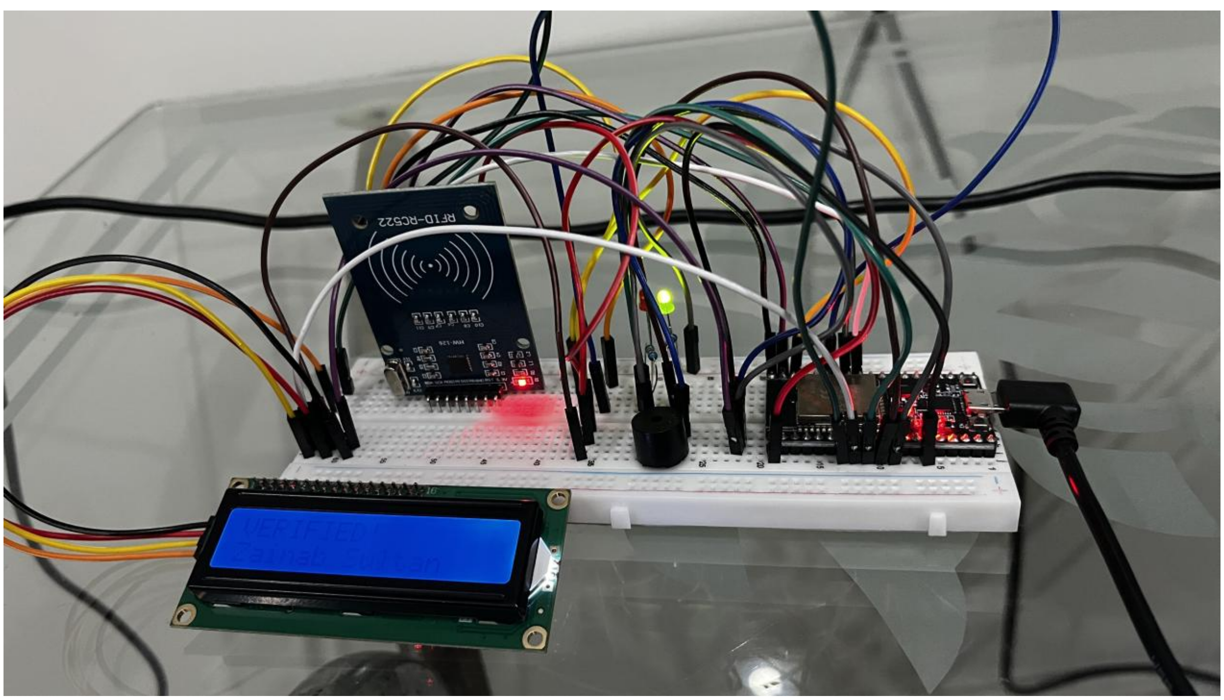
ESP32 successfully connects to Wi-Fi and RFID reader is ready to scan card. LED displays "Scan Your Card".

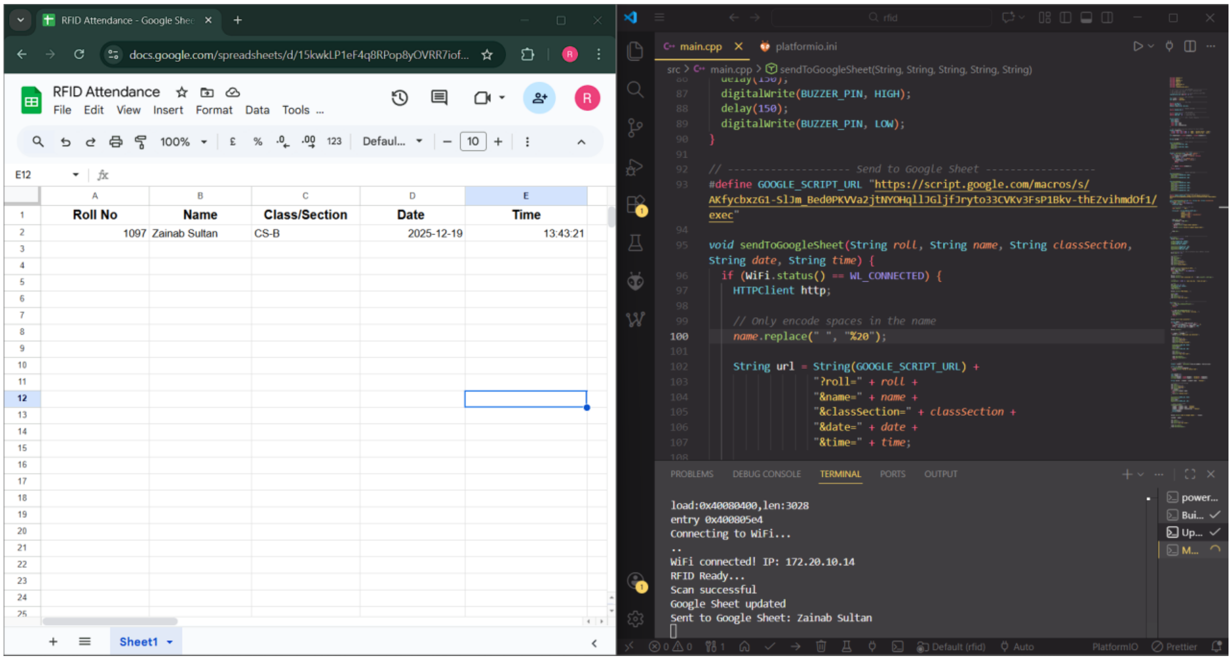




6.2 Valid Scan

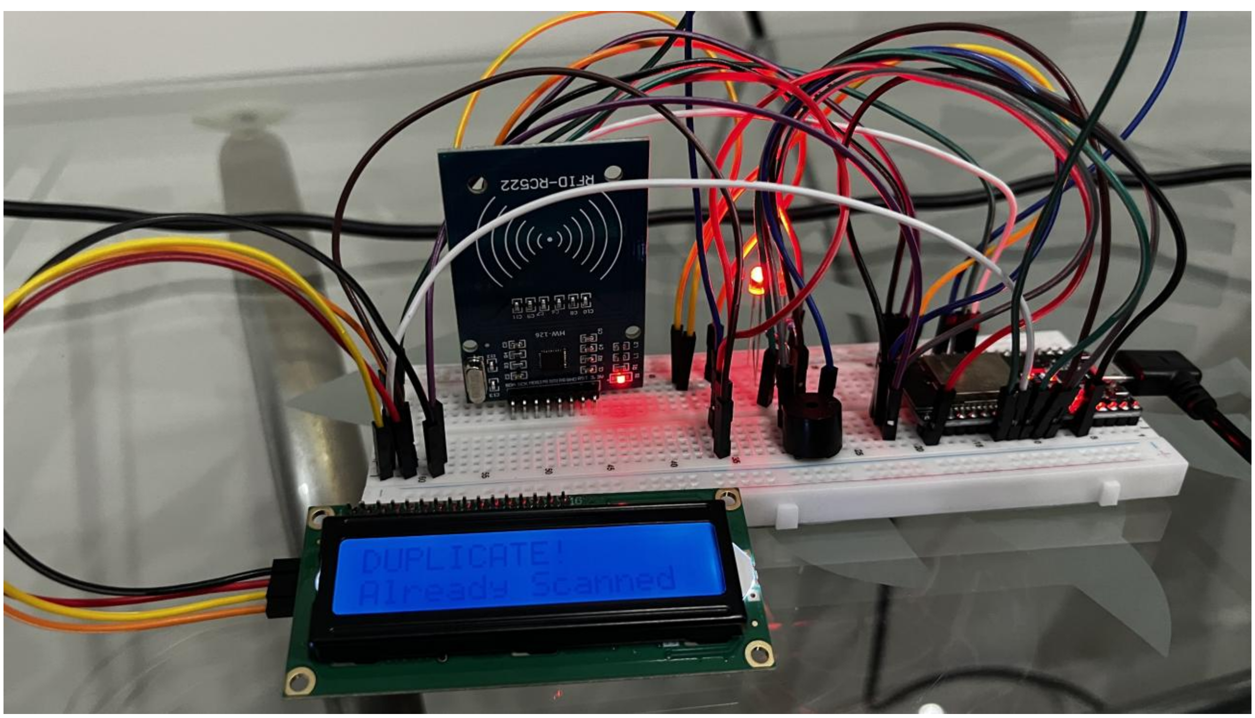
When an RFID card is scanned, RFID reader scans its UID and verifies corresponding student details. It makes sure that card is not scanned twice by comparing with last scanned UIDs. Once verified, Green LED blinks, Buzzer makes a short beep and the LED Displays "VERIFIED" with student's name. Attendance is marked on Google Sheets successfully.

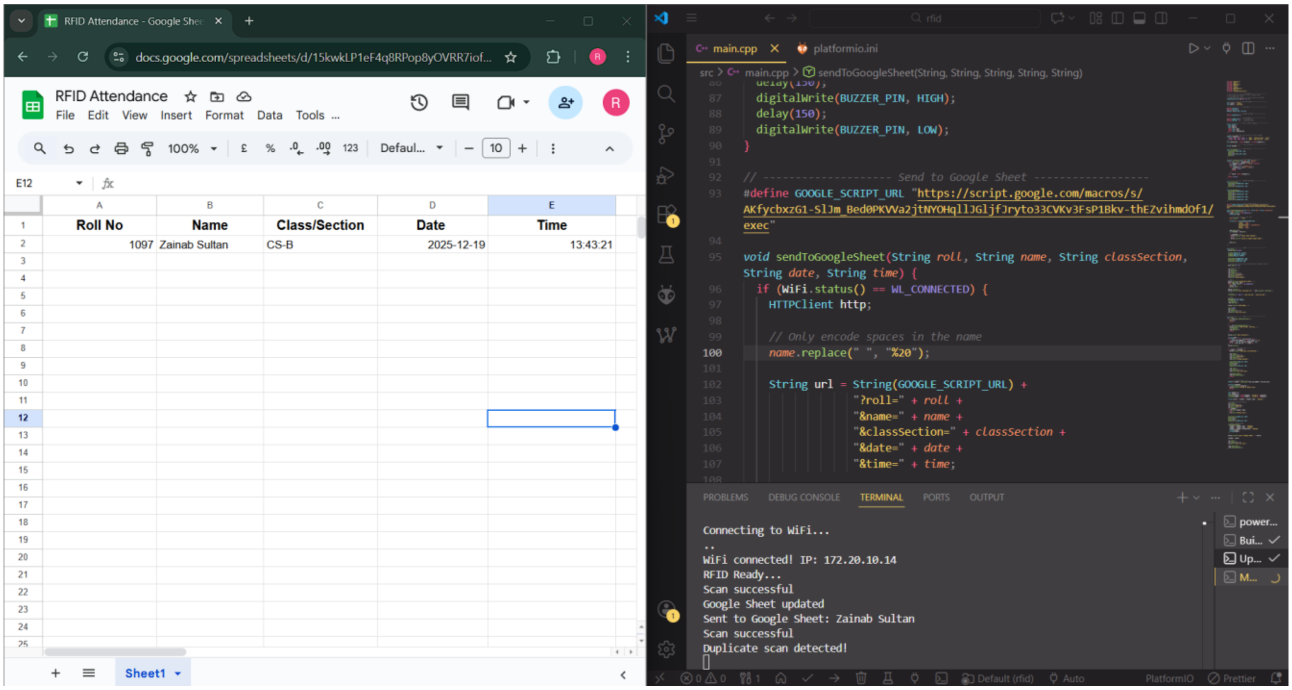




6.3 Duplicate Scan

If the card has been scanned before then Red LED blinks, Buzzer makes a double beep and the LED Displays “DUPLICATE! Already Scanned”. Attendance is not marked.





7. Future Enhancements

Mobile Application Integration: a mobile application can be developed for teachers and administrators to view attendance and generate reports conveniently from their smartphones.

Multi-Class/Section Support: the system can be enhanced to handle multiple classes/sections and subjects.

8. Conclusion

The RFID attendance system successfully automates attendance taking with a fast and accurate system. Use of RFID technology helps minimize human errors and attendance records stored on the cloud can be accessed from anywhere.